

CSE Alpha Engine

Design system, screen architecture and interaction model
for a calm, high-information investment platform

Companion to	CSE Alpha Engine — Master Specification v2.0
Design brief	Clean. Simple to navigate. Deep information rendered so an ordinary person can act on it. A psychologically non-triggering colour palette. Nothing that manufactures urgency.
Core tension resolved	Maximum information density with minimum arousal. These are usually traded off against each other; this document is mostly about how not to.
Stack	React · Tailwind · Recharts. All tokens supplied as CSS custom properties and a Tailwind config in §16.
Date	9 August 2026

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1 • Design Philosophy

Most financial interfaces are designed to make you trade. This one is designed to make you think, and then to get out of the way. Every choice below follows from that single inversion.

The analogy that governs the whole design

This is a **library reading room**, not a trading floor. A trading floor is loud on purpose: flashing prices, red and green everywhere, alarms, a sense that something is always happening that you might miss. That environment is optimised for reaction speed, and reaction speed is exactly the wrong instinct for a 12–36 month strategy in a market where a round trip costs you 3–5% before you have made anything. A reading room is quiet, well lit, dense with information, and organised so you can find the one thing you came for and then sit with it. Everything below is an attempt to build a reading room that happens to update itself.

The five laws

#	Law	What it means in the interface
1	Calm by default	No flashing, no live-ticking numbers, no countdown timers, no red alert banners, no notification badges on things that are not urgent. Prices update on a gentle fade, not a flash. The only genuinely time-sensitive event in this system is a watchlist name crossing its buy-below price, and it is the only thing allowed to interrupt.
2	Density without noise	The user asked for a lot of detail shown simply. The answer is not less information — it is <i>hierarchy</i> . One number is large; the rest support it. Every screen has exactly one primary answer, and everything else is available within one interaction of it.
3	Every number is a door	Any figure can be clicked to reveal how it was computed, from what inputs, from which source document, at what date. Three clicks from a composite score to a page of an audited PDF. Nothing is a dead end.
4	Show uncertainty, don't hide it	Ranges, not points. Dispersion bands. Provenance chips. Data-completeness meters. A confident-looking number that is actually a guess is the most dangerous object a financial interface can display.
5	Present, don't instruct	There is no BUY button and no verdict word anywhere in the product. There are zones, evidence and a decision form. The system's job is to make disagreement between models impossible to miss; the judgement is yours.

2 • The Colour System

2.1 Why the standard financial palette is a problem

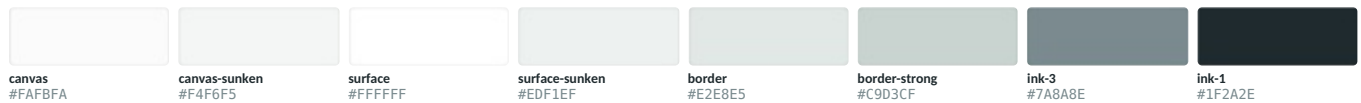
Saturated red and green is the default in every market product, and it is a poor choice for a long-horizon tool for three reasons:

- **Red is a threat signal.** High-saturation red is the strongest colour cue humans have for danger. Rendering a -2% day in alarm red recruits the same response as a warning light, and loss-aversion is already the dominant bias in investing. The interface should not amplify it.
- **The asymmetry cuts both ways.** Bright green makes gains feel like wins and encourages holding on to what is already expensive. If gains look euphoric and losses look alarming, the interface is quietly arguing for more trading.
- **It fails for roughly 8% of men.** Red-green is the most common form of colour vision deficiency. Any interface that carries meaning in that one axis is inaccessible by construction.

2.2 The palette

Low-saturation, warm-neutral, paper-based. Directional colour is present — you still need to see direction instantly — but it is *quiet*, and it is never the only carrier of meaning.

Neutrals — the paper



Canvas is a barely-warm off-white rather than pure white — it reduces glare over long reading sessions without looking sepia. Primary text is a deep slate rather than black; pure black on pure white is the highest-contrast combination available and it is fatiguing at this information density.

Brand — deep teal



Directional — quiet, and never alone



THE SUBSTITUTIONS AND WHY

Pine (#4E7C6A) replaces green. Desaturated enough that a good day looks like a fact rather than a celebration. **Clay (#A8674F) replaces red.** Warm, earthy, unmistakably "attention" but not "danger" — it is the colour of a terracotta pot, not a fire alarm. Pine and clay differ in hue *and* in lightness, so they remain distinguishable under every common form of colour vision deficiency and in greyscale. **Ochre (#B08A3E) is reserved exclusively for data quality and system state** — staleness, missing inputs, quarantine — and never for market movement, so that "something is wrong with the data" is never confused with "something is wrong with the company."

Categorical — charts and segments



Ordered by perceptual distance so that a two-series chart uses maximally separated colours, and an eight-series chart still resolves. All eight sit within a narrow lightness band so no single series dominates by brightness alone.

2.3 Semantic tokens — never use a raw hex in a component

Token	Value	Used for
--zone-strong-accumulate	#3F6E5A	Deepest buy zone on the price ladder
--zone-accumulate	#6E9B8E	Accumulate band
--zone-fair	#C2CDC7	Fair-value band — deliberately the least visually interesting zone
--zone-trim	#C9A15E	Trim band
--zone-exit	#A8674F	Exit band
--regime-risk-on	#4E7C6A	Macro regime indicator
--regime-transition	#B08A3E	Macro regime indicator
--regime-risk-off	#7E8F92	Risk-Off is <i>slate</i> , <i>not red</i> — it is a season, not an emergency
--prov-reported	#2C5F5D	Provenance chip: Reported
--prov-derived	#6E9B8E	Provenance chip: Derived
--prov-estimated	#B08A3E	Provenance chip: Estimated / Forecast
--prov-ai	#8A6E8F	Provenance chip: AI-assisted, unconfirmed
--prov-missing	#A7B3B5	Provenance chip: Unavailable
--thesis-intact	#4E7C6A	Thesis-drift verdict
--thesis-weakening	#B08A3E	Thesis-drift verdict
--thesis-broken	#A8674F	Thesis-drift verdict

DELIBERATE CHOICE WORTH DEFENDING

Risk-Off is rendered in **slate grey**, **not red**. A defensive macro regime is a normal, recurring condition of markets — roughly a third of all time. Rendering it as an emergency every time it appears would train you to dread it, and dread produces exactly the wrong behaviour, which is doing something. Grey says: conditions are unfavourable, own less, wait. That is the correct emotional register.

2.4 Dark mode

canvas
#12181A

surface
#1A2225

sunken
#151C1E

border
#2C3739

ink-1
#E4EAE8

brand-300
#7FB3A5

positive
#6FA891

negative
#C98A70

Never pure black. #12181A retains a slight blue-green cast that keeps the interface feeling like the same product, and avoids the halation that pure-black backgrounds cause with light text on OLED displays. Directional colours are lightened rather than saturated, preserving the low-arousal character.

2.5 Contrast and colour-blind verification

Pairing	Ratio	Standard
ink-1 on canvas	13.4 : 1	Passes AAA for body text
ink-2 on canvas	7.1 : 1	Passes AAA for body text
ink-3 on canvas	4.0 : 1	Large text and non-essential labels only
positive on canvas	4.8 : 1	Passes AA
negative on canvas	4.6 : 1	Passes AA
brand-500 on canvas	6.9 : 1	Passes AA at all sizes, AAA at large

Redundant encoding is mandatory. Direction is always carried by three signals at once: hue, lightness, and a glyph (▲▼—). Zones on the price ladder are additionally distinguished by position and by a text label. Chart series carry direct labels rather than relying on a colour legend wherever space allows. Every one of these views remains readable in full greyscale, and that is the acceptance test.

3 • Typography

Role	Face	Notes
Interface and body	Inter (fallback: system-ui)	Neutral, excellent at small sizes, extensive numeric features
All figures	Inter with <code>font-variant-numeric: tabular-nums</code>	Non-negotiable. Tabular figures make columns of numbers align, which is the difference between a table you can scan and one you must read.
Code, tickers, formulae	JetBrains Mono (fallback: ui-monospace)	Tickers are rendered monospaced so <code>HNB.N00000</code> is instantly recognisable as an identifier rather than prose

Step	Size / line	Weight	Use
display	34 / 40	600	The single hero number on a screen — fair value, composite score
h1	24 / 30	600	Screen title
h2	18 / 24	600	Section
h3	15 / 21	600	Card title
body	14 / 21	400	Prose, research notes
data	13 / 18	500	Table cells, metric values
label	11 / 15	600, +0.04em tracking, uppercase	Field labels, column headers
caption	11 / 15	400	Provenance, timestamps, footnotes

Three weights only — 400, 500, 600. No 700, no italics except for genuine emphasis in research prose. Restricting the weight palette is what stops a dense interface from becoming visually shouty as it accumulates features.

Measure. Research notes and any prose block are capped at 68 characters per line regardless of viewport. A 1,600px-wide paragraph is unreadable no matter how good the type is.

4 • Space, Grid, Elevation, Motion

Spacing scale

A 4px base. Permitted values only: 4 • 8 • 12 • 16 • 24 • 32 • 48 • 64 . Nothing else. Arbitrary spacing is how interfaces slowly lose their rhythm.

Grid

Breakpoint	Width	Columns	Behaviour
Desktop wide	≥1440px	12	Max content width 1360px, centred. Persistent left navigation rail 240px.
Desktop	1024–1439px	12	Navigation rail collapses to 64px icon rail
Tablet	768–1023px	8	Rail becomes a drawer; two-column cards become one
Mobile	<768px	4	Bottom tab bar; see §15.3

Elevation — restrained

Three levels total. Depth is signalled mostly by **border and background**, not by shadow. Heavy shadows read as "app chrome" and add visual noise at high information density.

Level	Treatment	Use
flat	Background change only	Table rows, sunken panels, inline sections
raised	1px border + 0 1px 2px rgba(31,42,46,.05)	Cards
floating	1px border + 0 8px 24px rgba(31,42,46,.12)	Popovers, drill-down panels, modals

Radius: 4px on inputs and chips, 6px on cards, 10px on modals. No pill-shaped buttons — they read as consumer app, and this is an instrument.

Motion

Transition	Duration / easing	Notes
Value update	400ms ease-out fade	A price that changes fades to the new value. It never flashes and never blinks.
Panel expand	180ms ease-out	Height and opacity
Route change	120ms opacity only	No slide, no page-turn
Chart draw	500ms once on mount	Never re-animates on data refresh — re-animation makes a stable series look volatile
Hover	90ms	Background and border only

All motion respects `prefers-reduced-motion` and drops to zero-duration. There is no motion anywhere in this product whose purpose is to attract attention.

5 • Data-Display Conventions

5.1 Numbers

Rule	Detail
Alignment	All numeric columns right-aligned, tabular figures, decimal points vertically aligned
Currency	LKR 21.40 · thousands separated · two decimals for prices, none for market cap
Large magnitudes	LKR 7.68tn · 412bn · 1.4bn · 860m — never a wall of digits. Full value on hover.
Percentages	One decimal. Percentage <i>points</i> written as pp to distinguish from percent change.
Ratios	One decimal, with the unit named: 14.2× P/E , not 14.2
Precision honesty	Fair value shown to the nearest rupee, never to two decimals. A valuation is not precise to the cent, and displaying it that way is a lie about certainty.
Ranges	Preferred over points wherever a range exists: LKR 268 — 341
Nulls	Rendered as Data unavailable in ink-3 with a reason on hover. Never a dash, never a zero, never omitted.

5.2 Direction — three signals, always

▲ +4.2%

positive

hue: pine

lightness: darker

glyph: up

— 0.0%

neutral

hue: slate

lightness: mid

glyph: dash

▼ -2.8%

negative

hue: clay

lightness: darker

glyph: down

Directional colour appears ONLY in:

· small text (13px and below)

· thin bars and sparkline strokes (max 3px)

· small chips (max 20px tall)

Directional colour NEVER appears as:

· a full-row background

· a full-width banner

· a card background

· anything larger than ~5% of the viewport area

That last constraint is the single most important rule in this document for keeping the interface calm. A table of forty rows where fifteen have clay-tinted backgrounds looks like a system failure. The same table with fifteen small clay figures looks like a Tuesday.

5.3 Provenance chips

Every displayed figure carries one. 16px tall, 2px radius, letter glyph plus tooltip.

Chip	Meaning	Tooltip content
R	Reported	Source document, page, filing date, "first available to market on..."
D	Derived	The formula, its version hash, and links to each input
N	Normalised	Itemised list of adjustments, plus the original unadjusted value
E	Estimated	Method used, peer set, confidence band
F	Forecast	Assumption set, scenario, and an "edit assumptions" link
AI	AI-assisted	The source text snippet, and a Confirm button. Until confirmed, the figure is excluded from every valuation and the chip is visible at all times.
—	Unavailable	Why it is missing and what would be needed to obtain it

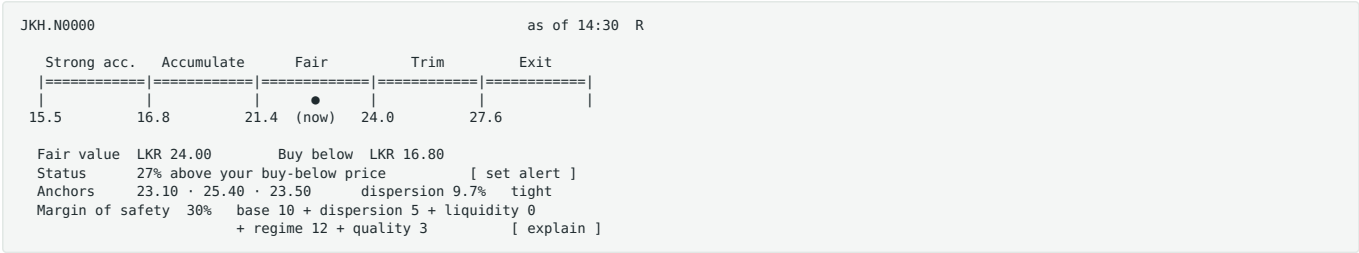
Aggregate indicator. Every company card and every ranking row carries a **data-completeness meter** — a 40px bar showing what share of required inputs exist. A stock scoring 88 on 54% completeness must look visibly different from one scoring 84 on 97%, before you click anything.

5.4 Timestamps

Every panel carries a quiet **as of** line. When data is older than expected, that line turns ochre and states the age plainly: *"Prices as of 14:30 yesterday — feed unavailable since 09:12."* Stale data is never silently rendered as current, and models that depend on it are visibly marked as not having recomputed.

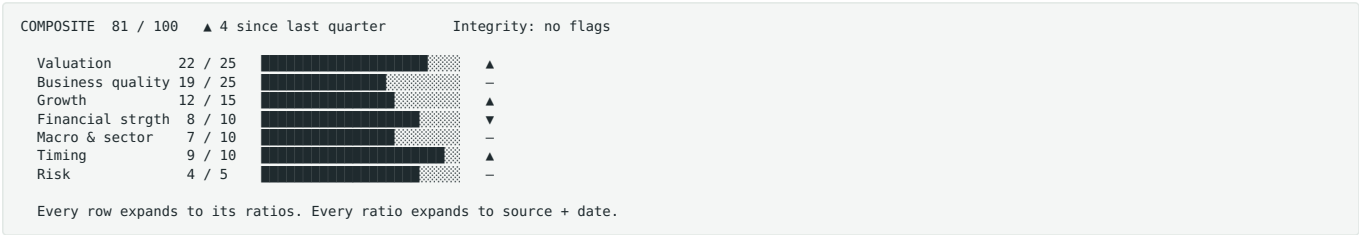
6 • Component Library

6.1 Price ladder — the signature component



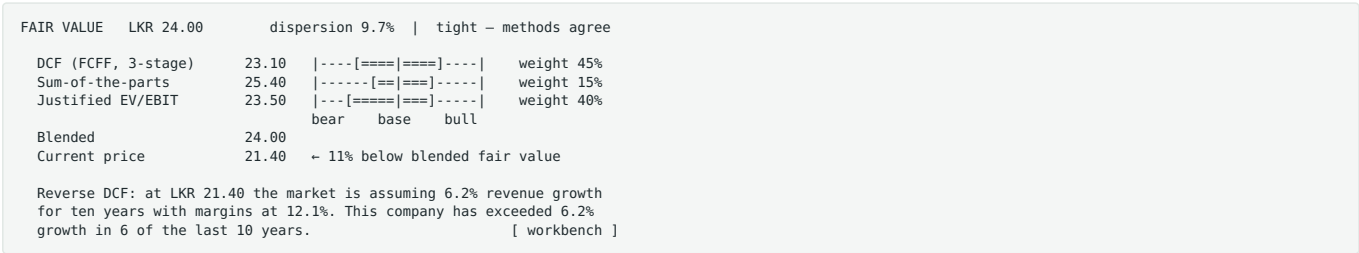
One horizontal bar, five zones by position and colour, the current price as a filled marker, and the gap stated in plain language beneath. The margin-of-safety breakdown is always visible in summary and expandable in full — because a user who cannot see *why* the buy-below price is where it is will eventually override it for the wrong reason.

6.2 Score breakdown



Bars use brand-300, not directional colour — a score is a position, not a movement. Only the trend arrow at the right carries direction.

6.3 Valuation anchor comparison



Each method shows its own bear-base-bull range as a bracket, so you can see not only that the methods disagree but *whether their ranges overlap*, which is a different and more useful question.

6.4 Remaining components

Component	Specification
Metric tile	Label, value, provenance chip, sparkline of the last 12 periods, trend arrow, sector percentile badge. Click opens the drill-down panel.
Trend cell	The table cell that replaces a bare ratio: value + inline 5-period sparkline + sector percentile. Turns a fact into a trajectory (Master Spec \$13) inside 90px.
Dispersion band	Horizontal band showing the min-max of valuation anchors with the blended value marked. Width is the message.
SOTP waterfall	Segment contributions ascending, then deductions descending, ending at value per share with the current price as a reference line. Each bar clickable to its own valuation.
Sensitivity tornado	Horizontal bars ranked by impact on fair value. Always visible on the valuation tab, never buried in an "advanced" section.
Regime gauge	Three-segment arc with a probability needle. Shows probability, never just the label. Risk-Off segment is slate.
Agreement indicator	Four small squares, one per model layer, filled when the layer supports the case and outlined when it does not. Four filled = easy decision. Two filled = the interesting case, and the card says so.
Thesis-drift strip	One line per holding: intact / weakening / broken, with the driving factor named. Sits at the top of the portfolio screen.
Evidence panel	Right-side slide-over, 480px, used for every drill-down. Never a modal — a modal blocks the context you are trying to compare against.
Decision form	Action, size, limit price, conviction 1–5, reasoning, and " what would prove me wrong? " — the last field is required and cannot be skipped.
Data-completeness meter	40 × 4px bar, brand-300 fill on border track, percentage on hover.
Ticker chip	Monospaced ticker + company short name + coverage-tier dot (Core / Watch / Excluded / Insufficient).

7 • Information Architecture

7.1 Navigation

TODAY	what needs my attention, and what is the weather
OPPORTUNITIES	the ranked board, the screener, the watchlist
COMPANIES	all ~286 names, searchable – every one has a file
PORTFOLIO	what I own, and whether the reasons still hold
MACRO	the regime, the variables, the project pipeline
JOURNAL	every decision I made and how it turned out
LAB	backtests and strategy variants (advanced)
DATA HEALTH	freshness, reconciliation, confirm queue (advanced)

Six primary destinations, two advanced ones separated by a rule. Global search (⌘K) resolves tickers, company names, metrics and screens, and is the fastest path to anything.

7.2 The two-minute rule

GOVERNING CONSTRAINT ON THE HOME SCREEN

On a normal day, **Today** must be fully readable in under two minutes and must usually conclude with "nothing to do." A 12–36 month strategy should not demand daily attention, and an interface that manufactures a reason to act every morning will push you into overtrading — which, in a market with 3–5% round-trip friction, is precisely how the cost structure destroys the returns the models worked to find. If the home screen routinely takes longer than two minutes, it is badly designed and must be cut back.

7.3 Depth model

Level 0	Today	one screen, four questions, two minutes
Level 1	Ranked board	20 rows, 8 columns, scannable
Level 2	Company file	one company, everything, tabbed
Level 3	Evidence panel	one number, fully explained
Level 4	Source document	the audited PDF, at the right page

Level 0 to Level 4 in four interactions, from anywhere. That is the whole navigation promise, and it is what "not a black box" means in practice.

8 • Screen 1 — Today

Four questions in descending order of importance. Sections 1 and 2 are the daily two minutes. Sections 3 and 4 are for a weekly sit-down.

Today		Friday 9 August 2026 · 08:41	
1. WHAT IS THE WEATHER?			
Regime TRANSITION P = 0.63 ▼ from Risk-On [gauge arc]			
Tilting toward Risk-Off: yields up, inflation above target, reserves drawn down, growth revised to 3%.			
Recommended gross exposure 58%		Your actual 82% ▲ over	
[HERO CHART – earnings yield minus 364-day T-bill spread, 5 years]			
Next events CBSL policy 30 Sep (52d) · CCPI 31 Aug (22d)			
IMF review · Budget			
2. WHAT NEEDS MY ATTENTION?			
● 2 watchlist names entered their accumulate zone			
● 1 holding fired an exit trigger – valuation			
● 3 new announcements on holdings		[triaged, 1 material]	
● 1 data quality alert – 1 ticker quarantined			
(on a normal day this section reads: nothing needs your attention)			
3. WHERE AM I?		4. WHAT IS ON THE BOARD?	
9 positions +14.2% YTD		Top 10 by composite, each with	
ASPI +6.8% S&P SL20 +5.1%		price, buy-below, gap %, agreement squares, days-to-	
Hit rate (20) 61% W/L 2.1x		build, coverage tier	
[factor exposure radar]			

The hero chart is the earnings-yield-minus-T-bill spread, not the ASPI. This is deliberate. The index tells you what happened; the spread tells you whether equities are cheap against the only alternative a Sri Lankan investor genuinely has. Putting it first trains the right instinct every single morning.

9 • Screen 2 — Opportunities

+-----+ Opportunities [Core 118] [Watch 74] [Excluded 41] [All 286] Filters sector ▾ archetype ▾ tier ▾ zone ▾ liquidity ▾ score ▾ ▣ hide names I cannot build in under 10 days +-----+									
#	Company	Score	Zone	Price	Buy blw	FV	Gap	Agree	
1	■ Company A	87	Accumulate	310	324	450	+45%	■■■■	
	Banks · Core	94%	build 4d						
2	■ Company B	82	Accumulate	215	207	280	+30%	■■■□	
	Diversified · Core	88%	build 7d		timing weak				
3	■ Company C	78	Fair	160	140	190	+19%	■■□□	
	Hotels · Core	76%	build 2d		macro adverse				
+-----+									
Diff since yesterday: 2 entered accumulate · 1 left · 4 scores moved >3									
+-----+									

Design decision	Reason
Coverage tier is a tab, not a hidden filter	Every listed company is analysed. Nothing is invisible — things are labelled. Excluded names are one click away with their exclusion reason on the row.
Two-line rows	Line 1 is the decision data; line 2 carries sector, tier, data-completeness meter, days-to-build and the single most important caveat. Cramming this into one line produces an unreadable 14-column table.
Agreement squares before gap %	A +45% gap where only two of four layers agree is a worse opportunity than +30% where all four do. Putting agreement adjacent to gap stops the eye anchoring on upside alone.
Build time is a first-class column	A great company you would need 40 sessions to accumulate is not an opportunity. The filter defaults to hiding these.
Persistent diff line	Answers "what changed and why?" without requiring you to remember yesterday's screen.
Zone as words, never a verdict	The column says <i>Accumulate</i> — a description of where the price sits relative to a computed band — not <i>BUY</i> .

10 • Screen 3 — Company File

One company, everything, seven tabs. This is the deepest screen and the one that must hold the most information without becoming a spreadsheet.

+-----+ █ JKH.N0000 John Keells Holdings PLC Core · Diversified holdings LKR 21.40 ▼ 0.5% mkt cap 312bn float 78% ADV 84m data 94% R +-----+ Overview Financials Valuation Sector fit Timing Risk Research +-----+ OVERVIEW +-----+ COMPOSITE 81 / 100 ▲4 [price ladder — \$6.1] [score breakdown — \$6.2] fair 24.00 buy below 16.80 +-----+ WHAT THIS COMPANY IS two sentences, plain language, no jargon THE CASE IN FIVE LINES + ROIC 14.2%, above its 11.0% cost of capital, and rising 3 years + Trades 34% below sum-of-the-parts, vs its own 5-year average of 21% + Net debt 0.8x EBITDA; no refinancing cliff inside 24 months - Leisure segment exposed to the tourism slowdown now in the data - Transition regime raises the required margin of safety by 12pp WHAT WOULD CHANGE THIS the falsification conditions, checkable AGREEMENT ███ fundamentals strong · valuation strong · macro neutral · timing weak. Historical base rate for this pattern: 54% hit rate, +11.2% average at 12m, 3.1 months of drawdown first. +-----+	
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Tab	Content
Overview	As above. Must be complete enough that a reasonable decision could be made from this tab alone.
Financials	Income statement, balance sheet, cash flow — 10 years, with every line carrying a provenance chip and a sparkline. Ratio panel grouped as profitability / growth / strength / efficiency / shareholder returns, each cell a trend cell (\$6.4). DuPont decomposition of ROE. Segment table where disclosed. Link to the filed PDF at the right page on every line.
Valuation	Anchor comparison (\$6.3), scenario table, Monte Carlo distribution with P(FV > price), sensitivity tornado, reverse DCF statement, SOTP waterfall for conglomerates, own-history multiple bands, justified-multiple scatter with this company marked, peer comparison table.
Sector fit	The archetype framework's specific metrics, peer ranking on each, sector sensitivity to the current regime, and the plain-sentence macro translation: "Oil +18% over three months; this company imports 62% of input cost with a two-quarter pass-through lag; modelled EBIT impact -180bp, reflected in the bear case." Related projects from the register.
Timing	Price chart with the 52-week high anchor and the entry band overlaid. Signal battery breakdown, branch label, technical panel across daily / weekly / monthly. Volume profile.
Risk	Integrity register with every flag and every clean check shown, Beneish / Sloan / Altman / F-Score, related-party exposure, FX exposure, liquidity profile and days-to-exit, factor loadings with the structural-break flag, residual volatility, concentration of revenue or counterparty.
Research	The AI note (Master Spec \$44), rendered with inline provenance chips, versioned and diffable against the previous quarter's note. A "what changed" toggle is the most valuable control on this tab.

Persistent footer on every tab: the decision form. You can record a decision from anywhere in the file without losing your place, and the model state is frozen at that instant.

11 • Screen 4 — Valuation Workbench

Where an ordinary person becomes an analyst. Every assumption is a slider or a field; the fair value recomputes live; and the system's base case is never overwritten.

Valuation workbench · JKH.N0000

[system base] [my scenario ▼]

ASSUMPTIONS		FAIR VALUE
Revenue growth Y1-5	6.2%	LKR 24.00 → LKR 26.80
system: 6.2 (3y CAGR adj.)		current price 21.40
EBIT margin	12.1%	upside +25%
Tax rate	28.0%	
Capex / revenue	4.1%	[live-updating anchor bars]
WACC	11.0%	
Terminal growth	4.5%	Sensitivity: WACC dominates.
		±100bp moves fair value ±18%.
[reset to system base]		

Every slider shows the system value, the peer median and this company's own 5-year range beneath the track, so you can see when you are being optimistic before the number tells you.

WHY THE GUARD RAILS MATTER

A workbench with unconstrained sliders is a machine for justifying a decision you already made. Showing the historical range beneath each control converts "I think growth will be 14%" into "I am assuming growth 8pp above anything this company has achieved in a decade," which is a much harder thing to do quietly. Scenarios you build are saved and attached to the decision record, so a post-mortem can later ask whether your assumptions or the system's were closer.

12 • Screens 5–7

Screen 5 — Portfolio

+-----+ Portfolio 9 positions · LKR 24.8m · +14.2% YTD · ASPI +6.8% +-----+						
THESIS STATUS						
● intact 6 ● weakening 2 ● broken 1						
Company D broken integrity flag raised – related-party receivables						
up from 18% to 34% of total. [review]						
Company E weakening fair value cut 12% on discount-rate change						
+-----+						
Holding Cost Now P&L Zone Thesis Nearest trigger						
...						
+-----+						
Allocation by sector · by archetype · by liquidity band						
Portfolio factor exposure vs target [radar] · gross vs recommended						
Contribution to risk by position · crowding · days-to-exit whole book						
+-----+						

Thesis status sits above P&L, on purpose. P&L is what happened; thesis status is what to do. Ordering them this way every day is a small, repeated nudge toward the right question.

Screen 6 — Macro

Regime gauge with probability · the hero spread chart · macro variable heatmap with z-scores against their own history · impulse-response panels · error-correction half-life stated in plain language ("*about 28% of the gap closes per month — roughly a 2.1 month half-life*") · sector sensitivity matrix · event calendar with historical cumulative-abnormal-return overlays · the national project register filtered to confirmed status. Every variable expands to its source, release date and revision history.

Screen 7 — Journal

The highest-value screen in the product over a three-year horizon, and it looks like the least exciting. Every signal, every decision, every override, logged with the reasoning as written at the time and the full model state frozen alongside. Filterable by outcome, conviction, regime, sector, agreement level. Includes the override analysis: returns on trades where you went against the composite versus with it. This is the only mechanism that distinguishes "the model was wrong" from "I overrode a correct model," and that distinction is what turns a tool into a skill.

13 • Screens 8–9

Screen 8 — Lab

Configure and run strategy variants. Equity curve with per-regime shading, gross versus net side by side, Deflated Sharpe, capacity curve, parameter sensitivity heatmap, baseline comparisons. **The net line is drawn solid and the gross line dashed** — a small choice that keeps attention on the only number that is real.

Screen 9 — Data health

Source freshness board, reconciliation results, quarantined tickers with reasons, and the **confirmation queue** — AI-extracted figures awaiting human promotion to Reported, each shown beside its source snippet with the page image. This queue is where data quality is actually maintained, so it deserves a real screen rather than an admin afterthought.

14 • The Explainability Pattern

One interaction pattern, used identically everywhere, so it becomes automatic.

Click any number

↓

Evidence panel slides in from the right (480px, context stays visible)

↓

WHAT IT IS

HOW IT IS BUILT

THE INPUTS

HOW IT COMPARES

WHY IT MOVED

SOURCE

one plain sentence, no jargon

the formula, rendered readably

each with its own value, provenance chip and date

this company's history · sector · market

decomposition of the change since last period

link to the filed document at the right page

The panel is recursive: any input inside it is itself clickable, pushing a breadcrumb. From a composite score you reach a pillar, then a ratio, then a line item, then the audited PDF page — four steps, one consistent gesture.

THE RULE THIS ENFORCES

You cannot sensibly override a system you cannot inspect, and a system you cannot override will eventually be abandoned. Explainability is not a transparency feature here; it is what makes the human-in-the-loop design viable at all.

15 • States, Accessibility, Mobile

15.1 Every component specifies six states

State	Treatment
Loading	Skeleton at the final layout's dimensions. Never a spinner over a collapsing layout.
Empty	Says what would fill it and what to do — <i>"No watchlist names have entered their accumulate zone. Add names from Opportunities."</i>
Partial	Renders what exists, marks what does not, and shows the completeness meter. Never all-or-nothing.
Stale	Ochre timestamp with the age and the reason. Dependent model outputs are visibly marked as not recomputed.
Error	What failed, what it affects, what still works, what happens next. Never a raw stack trace, never a bare "something went wrong."
Quarantined	Ticker-level. Full-width ochre notice on the company file, all model outputs suppressed rather than shown stale, with the reconciliation failure explained.

15.2 Accessibility

- WCAG 2.2 AA throughout; AAA on body text
- Full keyboard operability, visible 2px focus rings in brand-500, logical tab order, skip links
- All charts have an accessible data-table equivalent behind a toggle — not a hidden alt-text summary, the actual numbers
- ARIA live regions for value updates, set to `polite` so a screen reader is never interrupted mid-sentence by a price tick
- Colour is never the sole carrier of meaning, anywhere, verified by a greyscale screenshot test in CI
- Minimum touch target 44 × 44px; minimum text size 11px
- Respects `prefers-reduced-motion` and `prefers-color-scheme`

15.3 Mobile

Mobile is for **checking, not deciding**. Bottom tab bar: Today · Opportunities · Portfolio · Search. The company file collapses to a vertical accordion with Overview open by default. The valuation workbench, the lab and the data-health screen are **desktop only**, and the mobile build says so plainly rather than degrading them into something unusable. The decision form is available on mobile but requires the falsification field just as it does on desktop — no shortcuts because the screen is small.

16 • Design Tokens as Code

```
:root {
  /* neutrals */
  --canvas:#FAFBFA; --canvas-sunken:#F4F6F5;
  --surface:#FFFFFF; --surface-sunken:#EDF1EF;
  --border:#E2E8E5; --border-strong:#C9D3CF;
  --ink-1:#1F2A2E; --ink-2:#4A5A5E; --ink-3:#7A8A8E; --ink-4:#A7B3B5;

  /* brand */
  --brand-900:#143430; --brand-700:#1F4744; --brand-500:#2C5F5D;
  --brand-300:#6E9B8E; --brand-100:#DCE8E4; --brand-50:#F0F5F3;

  /* directional */
  --pos-strong:#3F6E5A; --pos:#4E7C6A; --pos-bg:#E8F0EB;
  --neg-strong:#96543D; --neg:#A8674F; --neg-bg:#F7EDE8;
  --neutral:#7A8A8E;
  --caution:#B08A3E; --caution-bg:#FAF4E6; /* data quality ONLY */

  /* categorical */
  --cat-1:#2C5F5D; --cat-2:#6E9B8E; --cat-3:#46577A; --cat-4:#8A6E8F;
  --cat-5:#A8674F; --cat-6:#B08A3E; --cat-7:#7E8F92; --cat-8:#C2CDC7;

  /* semantic */
  --zone-strong-accumulate:var(--pos-strong);
  --zone-accumulate:var(--brand-300);
  --zone-fair:#C2CDC7;
  --zone-trim:#C9A15E;
  --zone-exit:var(--neg);
  --regime-risk-on:var(--pos);
  --regime-transition:var(--caution);
  --regime-risk-off:#7E8F92; /* slate, never red */
  --prov-reported:var(--brand-500); --prov-derived:var(--brand-300);
  --prov-estimated:var(--caution); --prov-ai:#8A6E8F;
  --prov-missing:var(--ink-4);
  --thesis-intact:var(--pos); --thesis-weakening:var(--caution);
  --thesis-broken:var(--neg);

  /* type */
  --font-ui:'Inter',system-ui,sans-serif;
  --font-mono:'JetBrains Mono',ui-monospace,monospace;

  /* space */
  --s1:4px; --s2:8px; --s3:12px; --s4:16px;
  --s5:24px; --s6:32px; --s7:48px; --s8:64px;

  /* radius + elevation */
  --r-sm:4px; --r-md:6px; --r-lg:10px;
  --e-raised:0 1px 2px rgba(31,42,46,.05);
  --e-float:0 8px 24px rgba(31,42,46,.12);

  /* motion */
  --t-fast:90ms; --t-base:180ms; --t-value:400ms;
  --ease:cubic-bezier(.2,0,.2,1);
}

[data-theme="dark"] {
  --canvas:#12181A; --canvas-sunken:#151C1E;
  --surface:#1A2225; --surface-sunken:#151C1E;
  --border:#2C3739; --border-strong:#3C4A4D;
  --ink-1:#E4EAE8; --ink-2:#A8B6B8; --ink-3:#7E8F92; --ink-4:#5C6B6E;
  --brand-300:#7FB3A5; --brand-500:#3E8078;
  --pos:#6FA891; --pos-bg:#1C2E28;
  --neg:#C98A70; --neg-bg:#2E211C;
  --caution:#C9A65E;
}

/* mandatory global */
.num, table td, table th { font-variant-numeric: tabular-nums; }
@media (prefers-reduced-motion: reduce) { * { transition-duration:0ms !important;
  animation-duration:0ms !important; } }
```

17 • Anti-Patterns

Explicit prohibitions. These are the things a financial interface drifts toward if nobody writes them down.

Never	Because
Flashing or blinking price updates	Trains a reflex to react. Fade only.
Full-width red banners	Recruits a threat response for something that is usually routine.
Row backgrounds tinted by day's move	Turns a list into an emotional heat map. Direction belongs in small text.
A single BUY / SELL verdict	The product's purpose is to show evidence and disagreement, not to conclude.
Countdown timers or "act now" language	Manufactures urgency in a strategy whose holding period is measured in years.
Notification badges on non-urgent items	Badge fatigue destroys the one alert that matters.
Fair value to two decimal places	False precision is a lie about certainty.
Hiding missing data	Absence must be visible; a gap you cannot see is a gap you will assume away.
Charts without a zero baseline where one is meaningful	Truncated axes exaggerate movement.
Gamification — streaks, badges, confetti, leaderboards	Optimises engagement, and engagement is not the goal. Fewer, better decisions are.
Infinite scroll on a ranked list	Ranking implies a cut-off. Show the top N and say what N is.
Modal dialogs for explanations	Blocks the context you are comparing against. Slide-over panels only.
Placeholder or lorem content in any shipped state	In a financial product, a fake number that reaches a user once destroys trust permanently.
Auto-refresh that moves content under the cursor	Reading a table that reorders itself is impossible. Refresh on request, or on a stable key.

ONE-SENTENCE TEST FOR ANY FUTURE DESIGN DECISION

Does this help the user think more clearly, or does it help the product feel more exciting? If it is the second, it does not ship.

Companion to CSE Alpha Engine — Master Specification v2.0. Colour ratios stated in §2.5 are computed against the specified canvas token and should be re-verified in CI on any palette change. The greyscale screenshot test referenced in §15.2 is the acceptance gate for the redundant-encoding requirement. All tokens in §16 are the single source of truth; no component may reference a raw hex value.